

Questions

Lecture – 6

1. Draw the graphical representation of AC and DC.
2. Define form factor of AC.
3. The peak value of an alternating current is $20A$ and its frequency is $50Hz$. Find the root mean square value of the current. How long will take to reach the peak from the zero?
4. The amplitude of an AC source is $160V$ and its frequency is $60Hz$. If a resistance of 20Ω is connected with the source, find the effective voltage, effective current and loss of energy due to heat per time.
5. If the equation of the alternating current is $i = 50 \sin 628t$, then find (a) peak value (b) frequency and (c) *rms* value (d) average value of the current.
6. The alternating *emf* is expressed as $E = 200 \sin 100\pi t$. What is its amplitude, average value, rms value and frequency?